

Product Review and Rating Management System

DBMS Assignment

1. Objective

The Product Review and Rating Management System is designed to store customer feedback and ratings for products. It allows users to retrieve product review details, calculate average product ratings using aggregate functions, and identify highly rated products.

2. Create Review Table

```
CREATE TABLE Review (  
    Review_ID INT PRIMARY KEY,  
    Customer_ID INT,  
    Product_ID INT,  
    Review_Text VARCHAR(500),  
    Review_Date DATE  
);
```

3. Create Rating Table

```
CREATE TABLE Rating (  
    Rating_ID INT PRIMARY KEY,  
    Review_ID INT,  
    Product_ID INT,  
    Rating INT CHECK (Rating BETWEEN 1 AND 5),  
    FOREIGN KEY (Review_ID) REFERENCES Review(Review_ID)  
);
```

4. Insert Customer Feedback

```
INSERT INTO Review VALUES  
(1, 101, 1001, 'Good product and quality is excellent', '2026-09-01'),  
(2, 102, 1001, 'Very useful product', '2026-09-02'),  
(3, 103, 1002, 'Average quality', '2026-09-03'),  
(4, 104, 1003, 'Excellent product', '2026-09-04'),  
(5, 105, 1003, 'Worth the money', '2026-09-05');
```

5. Insert Ratings

```
INSERT INTO Rating VALUES  
(1, 1, 1001, 5),  
(2, 2, 1001, 4),  
(3, 3, 1002, 3),  
(4, 4, 1003, 5),  
(5, 5, 1003, 4);
```

6. Retrieve Product Review Details

```
SELECT  
    Review.Review_ID,  
    Review.Customer_ID,  
    Review.Product_ID,  
    Review.Review_Text,  
    Rating.Rating  
FROM Review
```

```
JOIN Rating
ON Review.Review_ID = Rating.Review_ID;
```

7. Calculate Average Product Rating

```
SELECT
    Product_ID,
    AVG(Rating) AS Average_Rating
FROM Rating
GROUP BY Product_ID;
```

8. Identify Highly Rated Products

```
SELECT
    Product_ID,
    AVG(Rating) AS Average_Rating
FROM Rating
GROUP BY Product_ID
HAVING AVG(Rating) >= 4;
```

9. Expected Average Rating Output

Product ID	Average Rating
1001	4.5
1002	3.0
1003	4.5

10. Highly Rated Products

Products with an average rating of 4 or above are considered highly rated. In this example, Product 1001 and Product 1003 have an average rating of 4.5.

11. Conclusion

The system efficiently manages product reviews and ratings. SQL commands such as CREATE TABLE, INSERT, JOIN, AVG(), GROUP BY, and HAVING are used to store, retrieve, analyze, and identify highly rated products.